

AIAA Conceptual Design Report

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Project Need

The project need is to design, build, and fly an unmanned aerial vehicle that abides by the rules and regulations of the American Institute of Aeronautics and Astronautics (AIAA) Design Build Fly competition utilizing the engineering design process until completion of the aircraft similar to the photos below of past AIAA designed aircrafts.



Figure 1: Example Aircraft fit for Competition [1]

Project Summary

Every year, the AIAA Foundation hosts a competition for university students around the world to design, manufacture, and compete with RC airplanes. The point of this competition is to replicate the process of designing full-scale airplanes to meet mission requirements with parameters given to the best of our abilities.

Our goal with this design is to build a plane satisfying constraints of the competition. Along with this, we want to build a plane that satisfies objectives and functions required to fly successfully and do well on competition missions. The constraints, objectives, and functions can be seen in the attributes list.

Attributes List

Physical Design & Structure

Characteristics	O	C	F
3-5-foot Wingspan		X	
TOGW < 55 lb		X	
One battery pack per prop.		X	
Max 100-watt hrs		X	
Assemble/Disassemble Capabilities (Modular)	X		X
Propeller Driven		X	

Flight Safety & Stability

Characteristics	O	C	F
Capable of landing and taking off			X
Stable flight	X		X
Electronically safe (arming fuse)	X	X	
Certifications: Pre-Tech and First-Flight	X		
TSA & DBF Safety Guidelines for UAV's		X	

Mission Performance & Payload Requirements

Characteristics	O	C	F
Cargo-Passenger Carry Capacity Ratio (1:3 min)		X	
Capable of holding many passengers and/or cargo			X
Complete 3 or more laps within a 5-minute window			X
Banner Deployment (10x2 in or larger)		X	X
DBF Mission Score Optimization	X		
Externally stowed banner		X	

Literature Review; Preliminary Research and Design Processes

Below is the following information regarding the weather conditions of this year's competition in Wichita Kansas for the month of April. Standard sea level conditions that will be taken into account will be the air pressure, air density, temperature all at the competition location listed below [2];

- Air Pressure: 14.7 psi
- Air Density: 0.0765 lbf/ft³
- Average Daily Temperature: 62°F
- Altitude: ~1300 ft above SL

This data above will be critical in the performance analysis of the chosen conceptual design so that the necessary calculations can be made to ensure that the aircraft can generate enough lift and for the weight analysis to be performed.

Literature Review; Performance Analysis Methods

Similar to the SPAROW team at University of Alabama Huntsville [3], and Journal of Engineering Science and Military Technologies [4], performance methods such as optimization codes and charts, MATLAB codes, and Excel spreadsheets aided in the selection of the conceptual design through a multitude of calculations and analysis. Utilizing these methods, a mathematical approach can be made to narrow down a chosen conceptual design. Figure 2 below shows a sample of an Excel performance analysis for the different aircraft designs.

Key			Pa (W)	V(ft/s)	CL	Cd	CL/Cd	Tr (lbf)	Tr/W	Ta (lbf)	Pr (W)	TE	Pe	R/C	MaxAngle
Power Available(Watts)	Pa	126500	126500	10	4.43E+00	1.07E+00	4.15	7.12E+02	0.24	12650	7.12E+03	1.26E+05	1.26E+06	426	88.66
Chord Length(in)	c	12	126500	20	1.11E+00	8.56E-02	12.95	2.28E+02	0.08	6325	4.56E+03	1.26E+05	2.53E+06	856	88.66
Wingspan(ft)	b	35.8	126500	30	4.93E-01	3.30E-02	14.95	1.97E+02	0.07	4217	5.92E+03	1.26E+05	3.79E+06	1284	88.66
Wing Area(ft^2)	S	174	126500	40	2.77E-01	2.41E-02	11.50	2.57E+02	0.09	3163	1.03E+04	1.26E+05	5.05E+06	1712	88.66
Aspect Ratio	AR	7.4	126500	50	1.77E-01	2.17E-02	8.18	3.61E+02	0.12	2530	1.80E+04	1.26E+05	6.31E+06	2138	88.66
Weight(lbf)	W	2950	126500	60	1.23E-01	2.08E-02	5.92	4.98E+02	0.17	2108	2.99E+04	1.26E+05	7.56E+06	2563	88.66
Density (lbf/ft^3)	p	0.076	126500	70	9.05E-02	2.04E-02	4.43	6.66E+02	0.23	1807	4.66E+04	1.26E+05	8.81E+06	2986	88.66
Banner Length (in)	Lb	40	126500	80	6.93E-02	2.03E-02	3.42	8.62E+02	0.29	1581	6.90E+04	1.26E+05	1.01E+07	3407	88.65
Rubber Duck	rb	1	126500	90	5.47E-02	2.02E-02	2.72	1.09E+03	0.37	1406	9.78E+04	1.25E+05	1.13E+07	3826	88.65
Hockey Pucks	h	1	126500	100	4.43E-02	2.01E-02	2.21	1.34E+03	0.45	1265	1.34E+05	1.25E+05	1.25E+07	4243	88.65
Oswald efficiency ratio	e	0.81	126500	110	3.66E-02	2.01E-02	1.83	1.62E+03	0.55	1150	1.78E+05	1.25E+05	1.37E+07	4657	88.65
Propulsion efficiency	k	0.053	126500	120	3.08E-02	2.01E-02	1.54	1.92E+03	0.65	1054	2.31E+05	1.25E+05	1.49E+07	5068	88.64
	np	0.8	126500	130	2.62E-02	2.00E-02	1.31	2.25E+03	0.76	973	2.93E+05	1.24E+05	1.62E+07	5475	88.64
	Cdo	0.02	126500	140	2.26E-02	2.00E-02	1.13	2.61E+03	0.89	904	3.66E+05	1.24E+05	1.73E+07	5879	88.64
CLmax (based on airfoil)	CLmax	1.2	126500	150	1.97E-02	2.00E-02	0.98	3.00E+03	1.02	843	4.50E+05	1.24E+05	1.85E+07	6280	88.63
Height of banner(in)	hb	2	126500	160	1.73E-02	2.00E-02	0.87	3.41E+03	1.16	791	5.45E+05	1.23E+05	1.97E+07	6676	88.63
Area of banner	Sb	80	126500	170	1.53E-02	2.00E-02	0.77	3.85E+03	1.30	744	6.54E+05	1.23E+05	2.09E+07	7068	88.62
Weight oz.		47200	126500	180	1.37E-02	2.00E-02	0.68	4.31E+03	1.46	703	7.76E+05	1.22E+05	2.20E+07	7455	88.62
WCL		20.6	126500	190	1.23E-02	2.00E-02	0.61	4.81E+03	1.63	666	9.13E+05	1.22E+05	2.31E+07	7838	88.61
W/S		17.0	126500	200	1.11E-02	2.00E-02	0.55	5.32E+03	1.80	633	1.06E+06	1.21E+05	2.42E+07	8215	88.61

Figure 2: Excel spreadsheet of Single Prop Performance Analysis

Design Space

Purpose: Identify a configuration satisfying the 2025-2026 AIAA DBF competition rules and constraints

Workflow:

- Began with broad brainstorming and then refined to four candidate concepts
- Morphological chart guided subsystem selection
- WCL analysis showed single-wing configurations had high stall speeds.
- Explored multi-wing options due to the improved lift and manufacturability balance [5]

Parameter	Description	Typical Range/Value
W	Aircraft Weight (lbf)	Estimate from previous DBF aircrafts
S	Wing Area (ft ²)	12 in, assuming a 1 ft chord length
b	Span (ft)	5 ft, max wingspan from rules
AR	Aspect ratio: Ratio of span to mean chord	Calculated from $AR = (b^2/S)$
e	Oswald Efficiency	0.7-0.9
C_{D0}	Parasite Drag Coefficient	0.045-0.06
C_{Lmax}	Maximum Lift Coefficient	1.1-1.4
P_A	Available Motor Power (W)	Based on similar motors from previous reports
η_p	Propellor Efficiency, Fraction of motor power converted to thrust	0.7-0.8

Performance Analysis

Most of the performance analysis is derived from common assumptions such as sea level conditions, Reynold's Numbers, etc. and parameters that signify the strengths and weaknesses of each conceptual design. Table 1 to the left shows these parameters that will be used in the performance analysis of each design.

Table 1: Common Aircraft Parameters

Performance Analysis Continued

Table 2 covers the necessary equations that are required for the performance analysis of the aircraft design that the group decided on. By understanding and solving for these equations, the appropriate dimensions can be calculated along with the proper power-plant requirements to power the aircraft.

Characteristic	Equation	Interpretation
Lift	$L = \frac{1}{2} * \rho * V^2 * S * C_L$	Amount of lift generated
Drag	$D = \frac{1}{2} * \rho * V^2 * S * C_D$	Amount of drag generated
Required Thrust	$T_R = \frac{W}{\frac{C_L}{C_D}}$	Thrust needed to maintain level flight
Available Thrust	$T_A = \frac{P_A * \eta_p}{V}$	The amount of thrust needed to determine operating regimes
Wing Loading	$\frac{W}{S}$	Lower value implies reduced stall speeds and shorter takeoff distance
Power Loading	$\frac{P}{W}$	Lower implies high endurance and efficiency
Thrust to Weight	$\frac{T}{W}$	Higher implies greater climb and acceleration capabilities
Required Power	$P_R = T_R * V$	Used to determine minimum and maximum flight speeds
Cubic Wing Loading	$WCL = \frac{W}{S^{1.5}}$	Lower value implies improvement in low-speed handling and controllability

Table 2: Necessary Equations for Performance Analysis

Conceptual Design 1

Strengths:

- Dual lifting surfaces.
- Modular fore- and aft- wing sections.
- Central Fuselage Payload Bay allowing balanced center-of-gravity placement.
- Standard control geometry.
- Banner release is located beneath the aft wing for unobstructed deployment.

Weaknesses:

- Slight downwash on the aft wing (mitigated with vertical offset and alt. angle of attack).

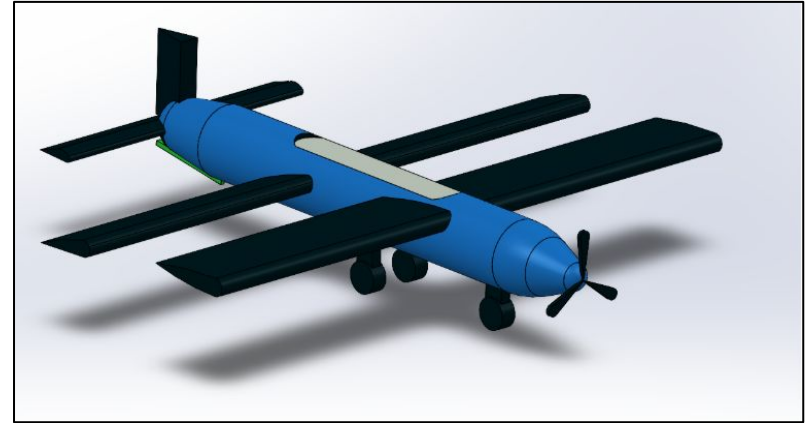


Figure 3: Conventional Tandem Wing design

Conceptual Design 2

Strengths:

- High lift in a short span.
- Rigid structure with interplane struts for stability.
- Compact, traditional configuration.
- Historically well studied

Weaknesses:

- Higher drag from wing interference and struts.
- Slightly heavier structure → lower aerodynamic efficiency.
- More complex to assemble.

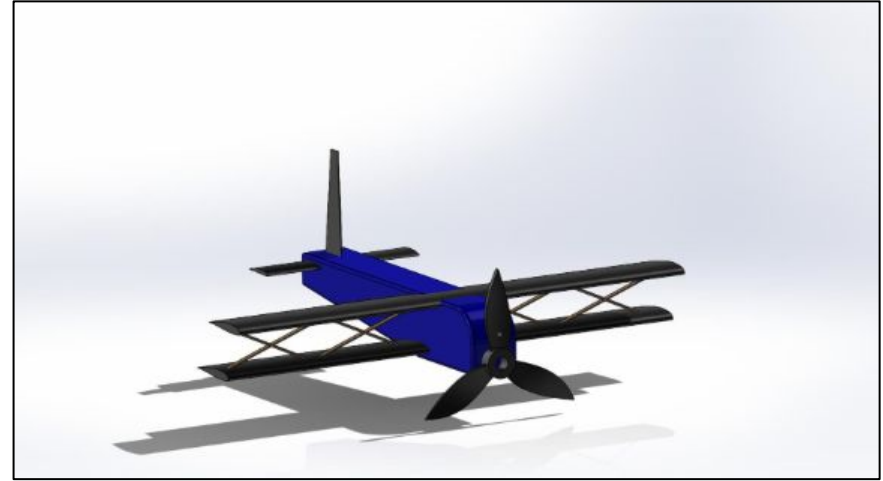


Figure 4: Bi-plane design

Conceptual Design 3

Strengths:

- Twin propulsion improves climb and redundancy.
- H-tail adds yaw stability.
- Maintains tandem-lift advantages with enhanced control.

Weaknesses:

- More complex structure and wiring.
- Slightly higher weight and cost.
- Taildragger gear less stable on the ground.

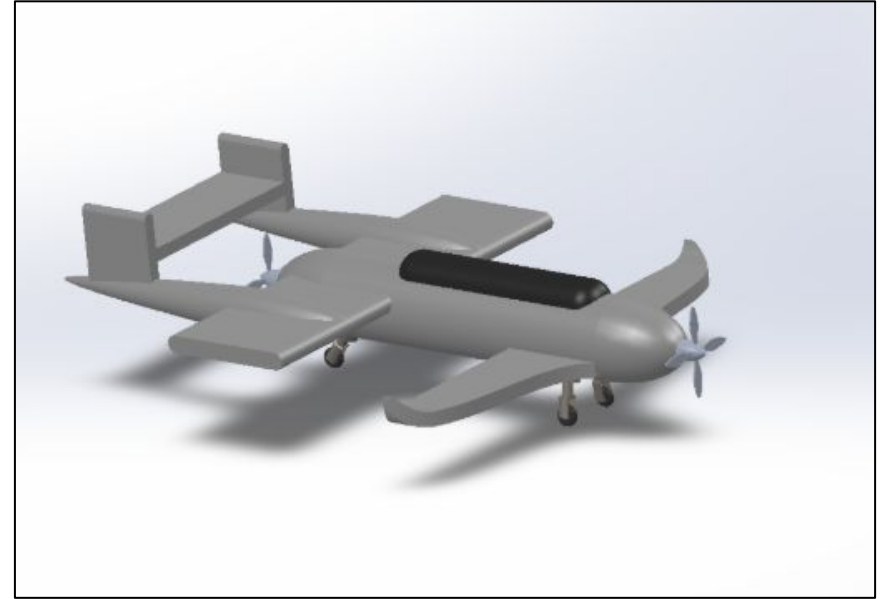


Figure 5: Dual Wing with H-tail design

Conceptual Design 4

Strengths:

- Simple, familiar layout — good structural stability.
- Dual motors improve redundancy.
- High wing enhances inherent static stability.

Weaknesses:

- Highest WCL and stall speed → poor low-speed performance.
- Requires more thrust and power → less endurance.
- Less efficient for heavy or high drag missions (like banner towing).
- Struts create additional parasite drag.

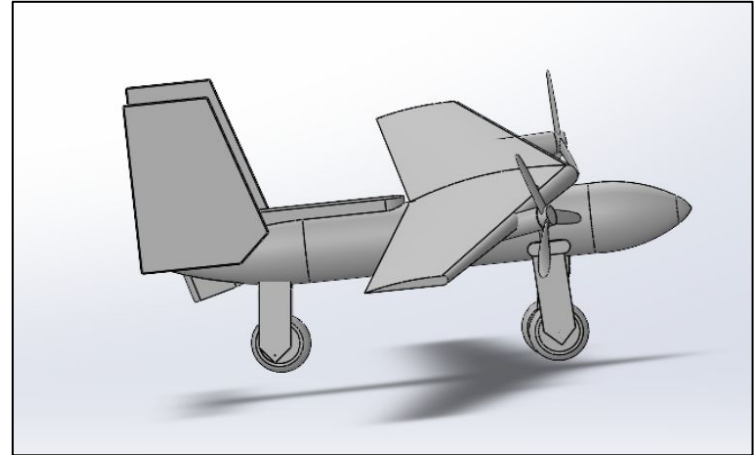


Figure 6: Single wing with H-tail design

Evaluation of Conceptual Designs

To objectively evaluate each conceptual design we needed a weighted decision matrix. This matrix is developed based on performance and manufacturing criterias. Key factors such as wing configuration, tail type, propulsion layout, landing gear etc. Each criterion was weight base on importance (1-10).

Criteria/Design Choice	Weight (1-10)	Evaluation (1-10)											
		Concept 1			Concept 2			Concept 3			Concept 4		
Wing Configuration	10	Tandem	9	90	Bi Plane	7	70	Tandem	9	90	High Wing	3	30
Tail Configuration	10	Conventional	10	100	Conventional	10	100	H-tail	9	90	H-tail	9	90
Landing Gear Configuration	7	Tricycle	10	70	Tricycle	10	70	Tail Dragger	8	56	Tricycle	9	63
Propulsion Systems	7	Single Prop	7	49	Single Prop	7	49	Dual Prop	10	70	Dual Prop	9	63
Cargo Access	8	Bottom side hatch	10	80	Bottom side hatch	10	80	Bottom rear hatch	8	64	Bottom rear hatch	8	64
Passenger Access	8	Top hatch	10	80	Top hatch	8	64	Top Hatch	10	80	Top Hatch	10	80
Manufacturability/Cost	10		10	100		8	80		9	90		10	100
Banner Mounting and System	7	Bottom of fuselage	9	63	Nose landing gear	6	42	Top rear of fuselage	10	70	Top rear of fuselage	10	70
Center of gravity and balance	10		10	100		9	90		10	100		8	80
Speed	8		9	72		8	64		10	80		7	56
				804			709			790			696

Figure 7: Decision Matrix of the Four Conceptual Designs

Evaluation of Conceptual Designs Results

Key Takeaways from Evaluation:

- The tandem-wing configurations (Concepts 1 & 3) clearly outperformed the other designs.
- Concept 1 achieved the highest total score (804); excellent aerodynamic efficiency and strong manufacturability.
- Concept 3 followed closely (790) with twin propulsion and H-tail benefits but added complexity.
- Concept 2 and Concept 4 underperformed due to higher drag and limited lift-to-weight efficiency.

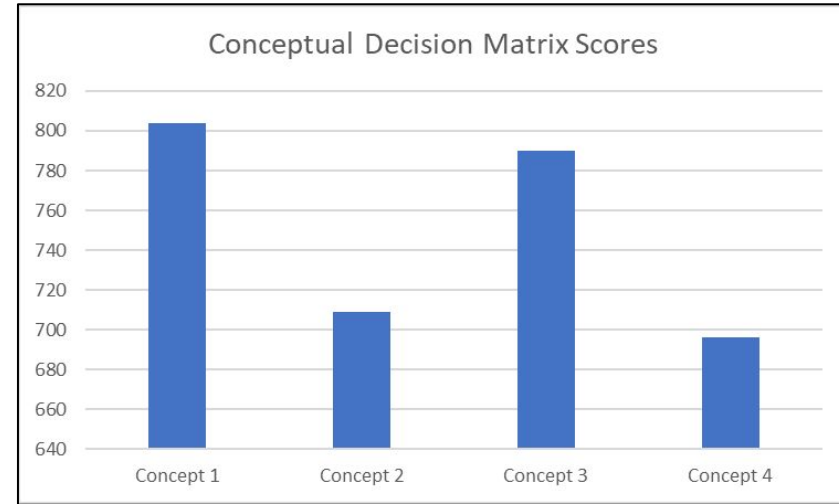


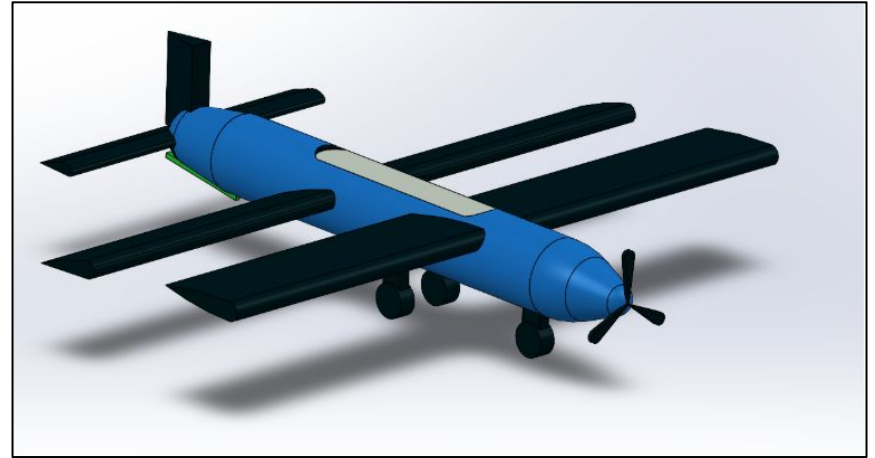
Figure 8: Decision Matrix Score Comparison

Final Design Conclusion

Lowest WCL and V_{stall} achieved → superior low-speed controllability and shorter takeoff distance. Validated by Weighted Decision Matrix; ranked highest overall among four alternatives.

Final Design Choice: Tandem wing with conventional tail

- Optimized lift efficiency
- Excellent controllability
- Ease of construction → practical for AIAA DBF competition



Codes and Standards

Listed below are Regulatory and Safety Frameworks found using AI that are significant to the completion of this project;

- **FAA Part 107:** Rules and regulations for small UAV operations.
- **FAA Order 8130.04:** Experimental UAV airworthiness certification requirements.
- **Title 49 USC §44809:** Recreational UAV operation and safety requirements.
- **AMA Safety Code:** Rules and regulations that ensure safe UAV flying practices.
- **NATO STANAG 4671:** Structural reliability, redundancy, and fail-safe systems, sets airworthiness standards amongst others in similar practices.

Manufacturing standards

- Minimize excess material and energy use.
- Avoid complex composites to cut emissions and cost.

Impacts

Our DBF aircraft design goes beyond competition, it reflects real-world aerospace engineering values and their broader implications on society, the environment, and industry.

Design Impact overview

- Broader implications across multiple areas.
 - Public Health , and safety.
 - Hands-on education,teamworks , and leadership .
- Cultural and global Impact.
 - Collaboration across cultures.
 - Shared innovation.
 - Sustainability awareness.

Sustainability,Environmental & Economic Implications

- Environmental
 - Lightweight design,efficient energy use, reduce waste.
- Global
 - Responsible sourcing and recyclable materials considerations.
- Economic
 - Balance cost and performance, eco-friendly

Coexistence of sustainability and performance in modern aircraft design.

Ethics

Our team followed the **National Society of Professional Engineers (NSPE) Code of Ethics** to ensure that all design, analysis, and testing activities were performed safely, responsibly, and professionally.

Public Health & Safety:

Designed the aircraft to minimize crash risks and meet all AIAA/FAA safety regulations. Safety of the public and participants is the top priority.

Client Obligations:

Maintained confidentiality and professionalism with advisors, the AIAA competition committee, and the University of Maine.

Technical Responsibility:

Each member contributed within their technical expertise and sought guidance when necessary to ensure sound engineering judgment.

Ethical Collaboration:

Promoted honesty, respect, and teamwork; all design decisions were supported by data and open communication.

Sustainability & Environmental Responsibility:

Used recyclable materials where possible, minimized waste, and focused on efficient manufacturing and reduced emissions.

References

- [1] “Design-build-fly 24-25,” UNC Charlotte, <https://crowdfund.charlotte.edu/project/40631> (accessed Oct. 19, 2025).
- [2] “Wichita, KS Weather in April: Average Temperature & Climate,” Wanderlog, <https://wanderlog.com/weather/58316/4/wichita-weather-in-april> (accessed Oct. 20, 2025).
- [3] Research Aircraft Optimization from Generated Requirements For Use In AIAA DBF Competition, <https://region2.aiaastudentconference.org/collection/2024/manuscripts/85176.pdf> (accessed Sep. 16, 2025).
- [4] “Modeling and Systematic Investigation of Small-Scale Propeller Selection,” Journal of Engineering Science and Military Technologies, <https://ejmtc.journals.ekb.eg/> (accessed Sep. 29, 2025).
- [5] “2025-26 Design, Build, Fly Rules,” AIAA, <https://aiaa.org/wp-content/uploads/2025/08/2026-DBF-Rules.pdf> (accessed Sep. 10, 2025).
- [6] “Code of Ethics for Engineers,” *National Society of Professional Engineers (NSPE)*, <https://www.nspe.org/career-growth/ethics/code-ethics> (accessed Oct. 22, 2025).

Questions?

